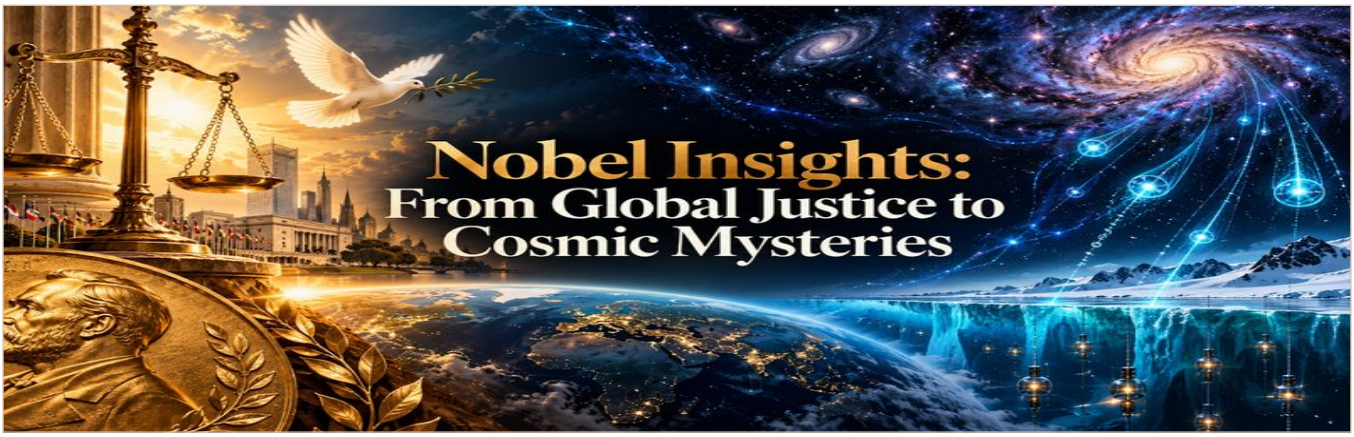


Akella Raghavendra Sir
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[GS PAPER II: INTERNATIONAL INSTITUTIONS • GS PAPER III: SCIENCE AND TECHNOLOGY • GS PAPER IV: ETHICS]
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Nobel Insights: From Global Justice to Cosmic Mysteries



Explore the rule of law, human rights and peace alongside IceCube's role in uncovering cosmic mysteries through neutrinos. Connect international justice and scientific discovery with key concepts for UPSC GS Papers II, III and IV.

Nobel Insights: From Global Justice to Cosmic Mysteries

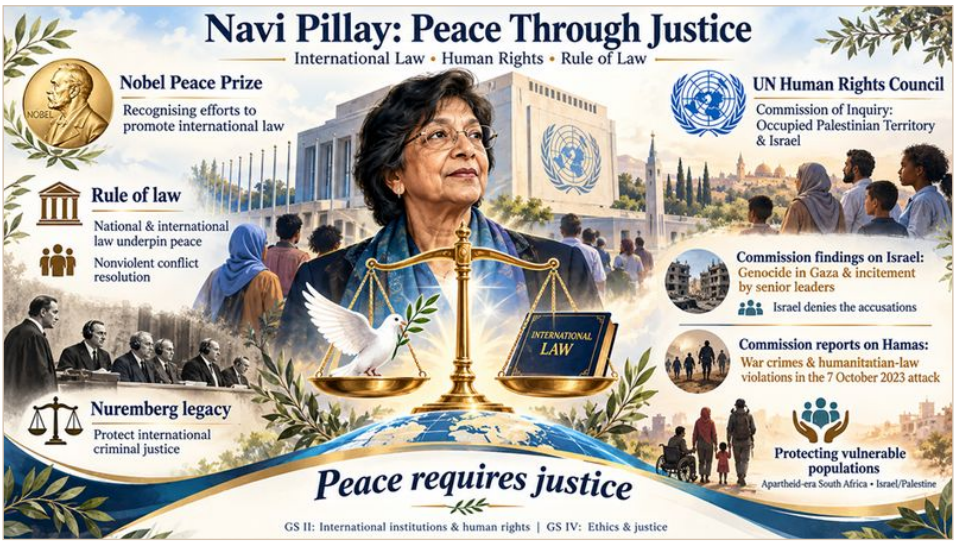
Syllabus areas:

- GS Paper II: International institutions, human rights, international law and conflict resolution.
- GS Paper III: Science and technology—IceCube, neutrinos and developments in particle physics.
- GS Paper IV: Ethics—justice, rule of law, moral courage and protection of human dignity.

Navi Pillay: International Law, Human Rights and the Pursuit of Peace through Justice

Navi Pillay was awarded the Nobel Peace Prize for her efforts to promote international law.

The Norwegian Nobel Committee emphasised that **national and international rule of law** underpin peace and the **nonviolent resolution of conflicts**. It highlighted that **peace requires justice** and that judges play a crucial role in maintaining order through courage, vision and determination.

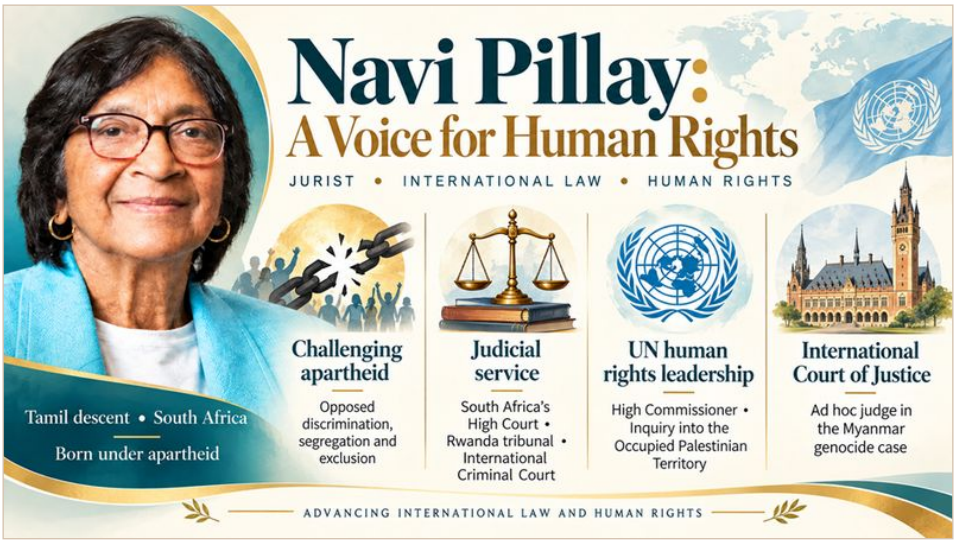


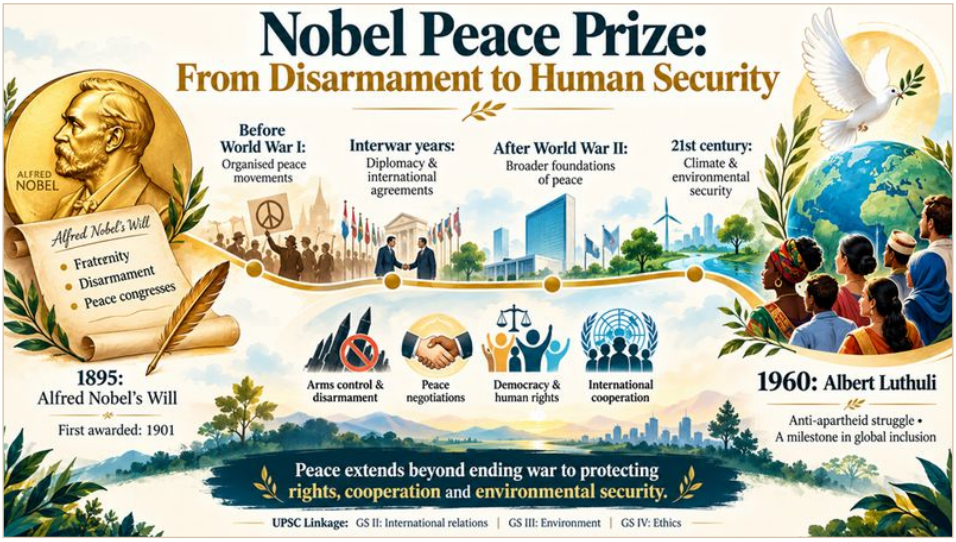
Navi Pillay addressed sensitive legal issues involving vulnerable populations in Israel and apartheid-era South Africa. She warned that weakening **international criminal justice** threatens the progress made since the **Nuremberg trials**, which helped lay the foundations of international law.

Pillay chaired the **Commission of Inquiry on the Occupied Palestinian Territory and Israel**, established by the **UN Human Rights Council**. The commission concluded that Israel was responsible for genocide in Gaza, with specific intent to destroy Palestinians there.

The commission led by Navi Pillay found that the Israeli President, Prime Minister and former Defence Minister had **incited genocide**. Israel denied these accusations.

Under her leadership, the commission also issued reports accusing **Hamas of war crimes and violations of international humanitarian law** during its **October 7, 2023, attack on Israel**.





Evolution of the Nobel Peace Prize:

1. **Original vision:** Alfred Nobel's will emphasised fraternity among nations, reduction of standing armies and promotion of peace congresses.
2. **Beginning in 1901:** The prize initially reflected a relatively narrow understanding of international peace.
3. **Early peace movements:** Before World War I, awards largely recognised pioneers of organised peace movements.
4. **Shift towards diplomacy:** During the interwar years, recognition increasingly went to political leaders pursuing peace through diplomacy and international agreements.
5. **Post-war expansion:** After World War II, the prize broadened to recognise several foundations of lasting peace.
6. **Disarmament and negotiation:** Arms control, disarmament and negotiated conflict resolution became major areas of recognition.
7. **Rights and global cooperation:** Democracy, human rights and efforts to build a better-organised international community gained prominence.
8. **Early geographical imbalance:** Until 1960, laureates were predominantly Western men; Argentina's Carlos Saavedra Lamas, awarded in 1936, was a notable exception.
9. **Growing global representation:** Albert Luthuli's 1960 award marked a turning point towards wider geographical inclusion, which accelerated during the 1980s.
10. **Environmental dimension:** In the 21st century, climate action and environmental protection became recognised as contributions to peace, reflecting a broader understanding of human security.



Terminology:

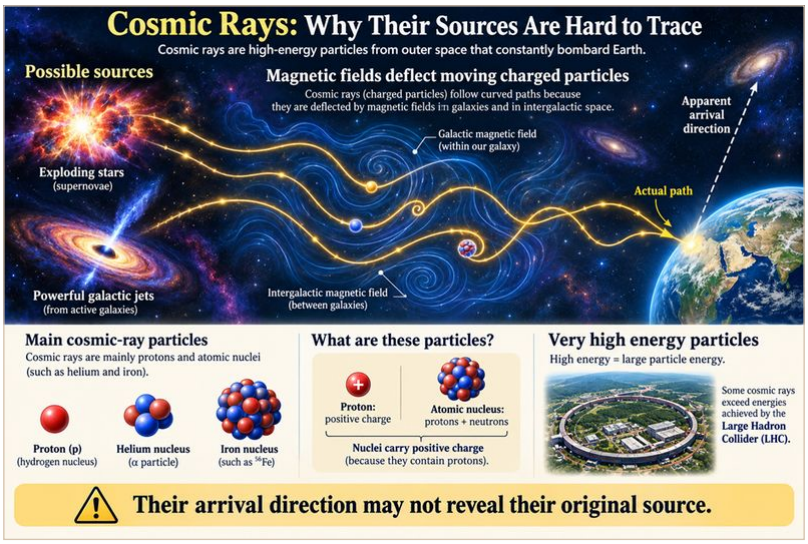
Term	Meaning
International Law	Rules and principles governing relations among states and other international actors.
Rule of Law	The principle that everyone, including governments, is subject to the law and accountable under it.
Human Rights	Fundamental rights inherent to every person, irrespective of nationality, religion, gender or other status.
International Criminal Justice	Mechanisms for investigating and prosecuting individuals responsible for genocide, war crimes and other serious international crimes.
International Humanitarian Law (IHL)	Rules applicable during armed conflicts that protect people not participating in hostilities and restrict methods of warfare.
War Crimes	Serious violations of international humanitarian law that entail individual criminal responsibility.
Genocide	Specified acts committed with the intent to destroy, wholly or partly, a national, ethnic, racial or religious group.
Incitement to Genocide	Direct and public encouragement to commit genocide; it is a punishable international crime.
Nuremberg Trials	Trials conducted after World War II to prosecute major Nazi leaders, helping establish individual accountability for international crimes.
Apartheid	A system of institutionalised racial segregation and discrimination formerly practised in South Africa.
UN Human Rights Council (UNHRC)	A United Nations intergovernmental body responsible for promoting and protecting human rights worldwide.
Commission of Inquiry	An investigative body established to examine alleged violations, establish facts and report findings.
Occupied Territory	Territory placed under the effective control of a hostile army.
Disarmament	The reduction or elimination of weapons and military capabilities.
Arms Control	Agreements or measures regulating the development, possession, deployment or use of weapons.
Human Security	An approach to security focused on protecting people's survival, livelihoods and dignity.

Ice Cube: A Quest for the Nobel Prize in Physics

The Belgian-American physicist **Francis Halzen** won the **2026 Nobel Prize in Physics** for helping build the **IceCube observatory in Antarctica**. IceCube uses thousands of sensors buried deep in the ice to detect subatomic particles called **neutrinos**.

Cosmic Rays and the Challenge of Identifying Their Sources

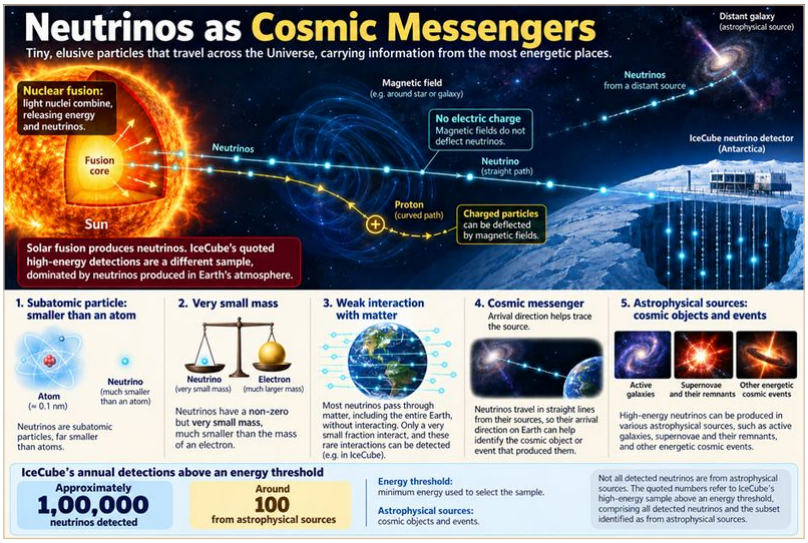
Cosmic rays are high-energy particles from outer space that constantly bombard Earth. They mainly consist of **protons and atomic nuclei**, including helium and iron. Some possess energies exceeding those achieved by the **Large Hadron Collider (LHC)**.



Exploding stars and powerful jets from distant galaxies are possible sources. However, cosmic rays are **electrically charged** and can be deflected by magnetic fields in space, making their arrival direction unreliable for identifying their origin.

Neutrinos as Cosmic Messengers

Neutrinos are subatomic particles with **no electric charge**, a **very small mass** and extremely weak interactions with matter. Nuclear reactions, including **fusion in the Sun**, produce them abundantly.

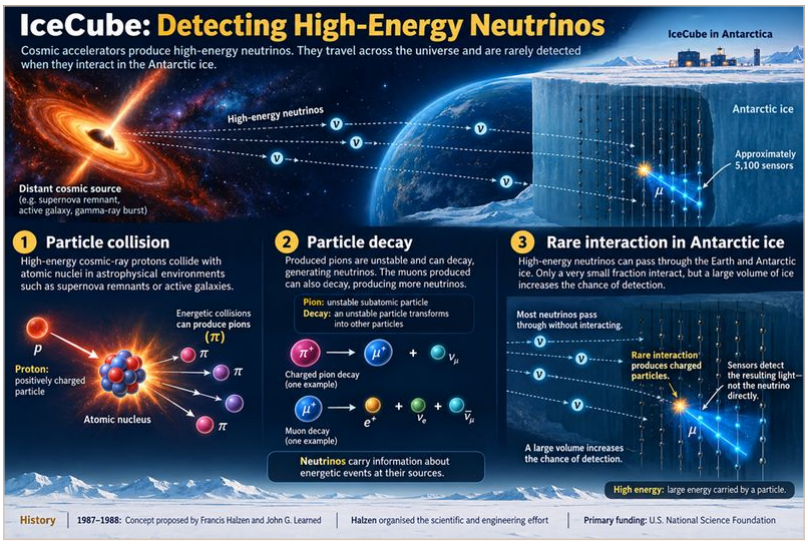


They travel through space without being deflected by magnetic fields, helping scientists trace their sources. Of approximately **one lakh neutrinos** IceCube detects annually above an energy threshold, around **100 originate from astrophysical sources**.

IceCube: Detecting High-Energy Neutrinos

When **cosmic-ray protons collide with atoms**, they can produce **pions**, whose decay can generate neutrinos. These high-energy neutrinos indicate energetic events at their sources.

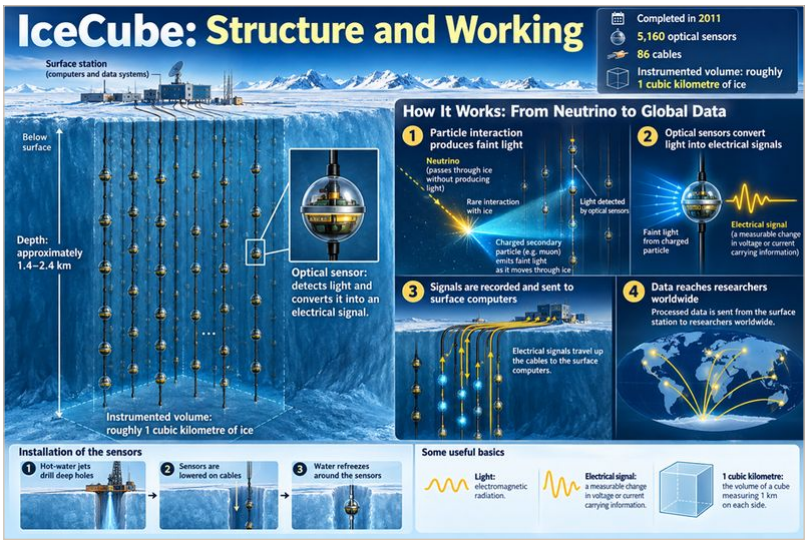
Because neutrinos rarely interact with matter, detection requires a large volume of material. **IceCube uses Antarctic ice embedded with approximately 5,100 sensors** to detect and study them.



Francis Halzen and John G. Learned proposed the concept in **1987–1988**. Halzen organised the long-term scientific and engineering effort, funded primarily by the **U.S. National Science Foundation**.

IceCube: Structure and Working

Completed in **2011**, IceCube consists of **5,160 optical sensors on 86 cables**, installed at depths of approximately **1.4–2.4 km** within Antarctic ice. It covers roughly **one cubic kilometre of ice**. Sensors were lowered into holes drilled using **hot-water jets** before the water refroze.

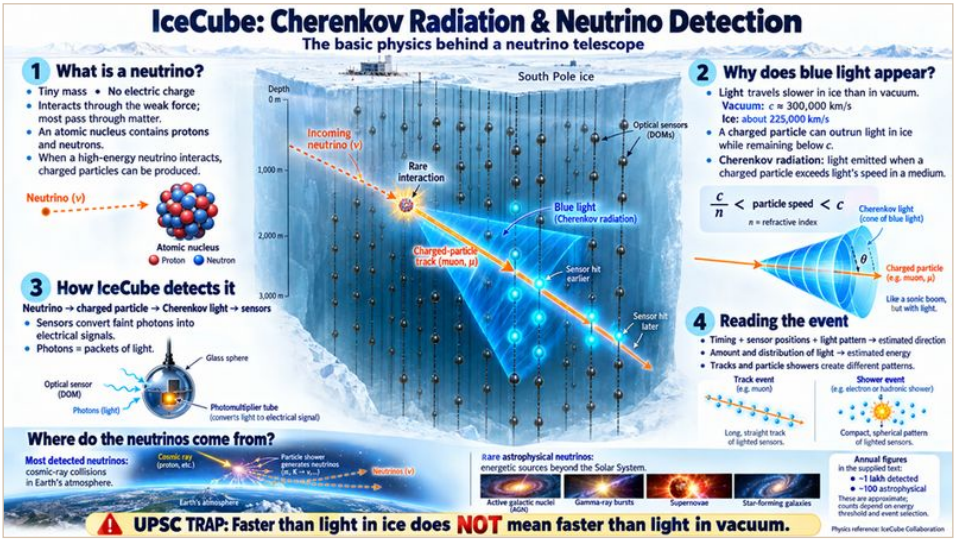


The sensors convert faint light produced during particle interactions into **electrical signals** and record them. Data is transmitted to surface computers and then to researchers worldwide.

IceCube: Cherenkov Radiation and Neutrino Detection

When a high-energy neutrino interacts with an atomic nucleus in the ice, it produces **charged particles**. Particles travelling faster than light travels **through ice** emit **Cherenkov radiation**, a faint blue glow detected by IceCube's sensors.

Scientists use the interaction pattern, sensor locations and timing of the light signals to reconstruct the neutrino's **energy and direction**. Of approximately **one lakh neutrinos** detected annually above an energy threshold, around **100 originate from astrophysical sources**.



IceCube: Key Discoveries in Neutrino Astronomy

- **2013:** Detected two neutrinos with energies around **140 times that of an LHC-accelerated proton**. Further analysis established a population of high-energy neutrinos originating beyond Earth, marking a milestone in **neutrino astronomy**.
- **2017:** Detected an energetic neutrino from a distant galaxy whose central black hole powered a radiation jet directed approximately towards Earth.
- **2022:** Identified an excess of neutrinos from **NGC 1068**, a galaxy approximately **46 million light-years away**, containing an active central black hole.

Scientists continue investigating the sources of the most energetic neutrinos and correlating IceCube detections with observations from other telescopes.

Terminology:

Term	Meaning in 30 words
IceCube Observatory	IceCube is a neutrino observatory in Antarctica, using thousands of optical sensors buried within ice to detect faint light from particle interactions and investigate energetic events occurring across the universe.
Cosmic Rays	Cosmic rays are energetic particles originating in space, mainly protons and atomic nuclei. Their electric charge allows magnetic fields to deflect them, making their original sources difficult to identify directly.
Neutrino	A neutrino is a subatomic particle with no electric charge and extremely small mass. It interacts weakly with matter, allowing it to pass through planets and reveal distant cosmic processes.
Magnetic Field	A magnetic field is a region where magnetic forces act. It can change the direction of moving charged particles, including cosmic rays, while electrically neutral neutrinos remain unaffected by it.
Nuclear Fusion	Nuclear fusion combines lighter atomic nuclei into heavier nuclei, releasing energy. It powers the Sun and other stars, producing neutrinos that help scientists investigate processes occurring within their stellar interiors.
Pion	A pion is an unstable subatomic particle produced during energetic interactions, including collisions involving cosmic rays. Charged pions decay into other particles, generating neutrinos that carry information about cosmic sources.
Cherenkov Radiation	Cherenkov radiation is light emitted when charged particles travel through a medium faster than light travels through that medium. IceCube detects this glow to study neutrino interactions within Antarctic ice.
Neutrino Astronomy	Neutrino astronomy studies celestial objects and energetic cosmic events by detecting neutrinos. Because neutrinos lack electric charge, their paths resist magnetic deflection, helping researchers trace their origins across the universe.

Term	Meaning in 30 words
Black Hole	A black hole is a region of spacetime where gravity prevents anything, including light, from escaping beyond its event horizon. Surrounding matter can release enormous energy as it falls inward.
Light-Year	A light-year measures distance, representing how far light travels through a vacuum in one year, approximately 9.46 trillion kilometres. Astronomers use it to express vast distances between objects in space.

Also read article on more Noble insights:Nobel Insights: Molecular Handedness to Brain's Light Switches